Online PAC Labeling from a Linear Programming Perspective

STATS 357 Project

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Abstract

We revisit cost-aware probably approximately correct (PAC) labeling through a linear-programming perspective. For a finite target pool, multiple cheap labeling sources, and access to an expert labeler, the oracle problem is to minimize labeling cost subject to an average-error constraint. The Lagrange dual of this LP gives a scalar shadow price for error, and the resulting itemwise minimizer is a fixed-price routing rule over cheap sources and expert deferral. Since the oracle losses are unavailable before labeling, the implementable method uses proxy losses only to generate candidate prices and policies. Finite-sample validity is then supplied by held-out calibration labels, which certify the actual loss of the selected policy without requiring the proxy to be correct. This separates optimization from inference: the LP and proxy model seek cost efficiency, while calibration provides the PAC certificate. We also describe an audited online extension in which randomized expert audits produce inverse-probability feedback for updating the dual price and controlling finite-horizon realized error.

1 Problem Setup and LP Formulation

1.1 Setup and Notation

There is a fixed pool of unlabeled items $X_1, \dots, X_T$, whose true labels $Y_1, \dots, Y_T$ are initially unknown, and the goal is to labeling the items correctly while minimizing labeling cost.

For each item t, we have access to k cheap ML models, which produce predicted labels $\widehat{Y}_t^{(j)}$. Also, the expert label Y_t is treated as correct but incurs an expansive query cost $c_{t,E} > 0$. In this project, we assume ML models are much cheaper than the expert, so we set $c_{t,j} = 0$ for all t and $j \in [k]$ and $c_{t,E} = 1$. The labeling loss is bounded:

$$0 \le \ell\big(Y_t, \widehat{Y}_t^{(j)}\big) \le 1, \qquad t \in [T],\ j \in [k]. \tag{1}$$

The expert action has zero labeling loss. Define the action set $\mathcal{A} = \{E, 1, \dots, k\}$, where E denotes deferral to the expert. For $a \in \mathcal{A}$, define the loss and cost for item t by

$$h_t(a) = \begin{cases} 0, & a = E, \\ \ell\big(Y_t, \widehat{Y}_t^{(a)}\big), & a \in [k], \end{cases} \qquad c_t(a) = \begin{cases} 1, & a = E, \\ 0, & a \in [k]. \end{cases} \tag{2}$$

A labeling rule chooses an action $a_t \in \mathcal{A}$ for each item. The resulting final label is

$$\widetilde{Y}_t = \begin{cases} Y_t, & a_t = E, \\ \widehat{Y}_t^{(a_t)}, & a_t \in [k]. \end{cases}$$

1.2 The Cost-Aware Labeling LP

We formulate the cost-aware PAC labeling problem as a linear program. Let $x_{t,a}$ be the probability of choosing action a on item t. If the true losses $h_t(a)$ were known, the optimal labeling policy would be the solution to the following LP:

$$\begin{aligned} \text{minimize}_x \quad & \frac{1}{T}\sum_{t=1}^{T}\sum_{a\in\mathcal{A}} c_t(a)x_{t,a} \\ \text{subject to} \quad & \frac{1}{T}\sum_{t=1}^{T}\sum_{a\in\mathcal{A}} h_t(a)x_{t,a} \le \varepsilon, \\ & \sum_{a\in\mathcal{A}} x_{t,a} = 1, \qquad t\in[T], \\ & x_{t,a} \ge 0, \qquad t\in[T],\ a\in\mathcal{A}. \end{aligned} \tag{3}$$

Intuitively, the LP minimizes labeling cost subject to an average-error constraint. The LP explicitly takes both the cost and the loss into account, and it models multi-source setting.

1.3 Lagrangian Dual and Shadow Price

The oracle LP (3) has a single global average-error constraint. Let $\lambda \ge 0$ denote the Lagrange multiplier, or shadow price, of the labeling error constraint. The dual problem is the one-dimensional concave maximization problem (See Appendix B.1 for details):

$$\max_{\lambda\ge 0} D(\lambda) := -\lambda\varepsilon + \frac{1}{T}\sum_{t=1}^{T}\min_{a\in\mathcal{A}}\{c_t(a) + \lambda h_t(a)\}. \tag{4}$$

This is the source of the fixed-price rule: once a price λ is fixed, the dual minimization separates across items, and the action for item t is given by the following simple rule:

$$a_t(\lambda) \in \arg\min_{a\in\mathcal{A}}\{c_t(a) + \lambda h_t(a)\}. \tag{5}$$

The dual price λ balances the trade-off between cost and error.

1.4 Roadmap

The LP perspective characterizes the optimal labeling policy by using a dual price λ. In practice, the true losses $h_t(a)$ are unknown before expert labels are observed, so the implementable method uses proxy losses $\widehat{h}_t(a)$ to generate candidate prices and policies. The finite-sample validity of the selected policy is then supplied by held-out calibration labels, which certify the actual loss of the selected policy without requiring the proxy to be correct. This separates optimization from inference: the LP and proxy model seek cost efficiency, while calibration provides the PAC certificate.

In the **offline setting**, we use proxy losses $\widehat{h}_t$ to generate fixed-price policies and reserve a calibration sample with expert labels to choose λ with a high-probability error certificate. The validity result does not rely on the proxy $\widehat{h}_t$ being correct, but it affects the labeling cost.

In the **online setting**, items arrive onlinely and labels are revealed only intermittently through random audits. Whenever the true label is observed, we obtain feedback about the realized loss h_t and update the price λ_t through an online dual update.

2 Offline PAC Labeling

In this section, we first define the proxy-price labeling policies and then use a labeled calibration set to select a certified price for the target pool.

2.1 Labeling with Proxy Losses

For a candidate price λ, the oracle action $a_t(\lambda)$ from (5) cannot be computed because $h_t(a)$ is unknown. We therefore choose actions by replacing $h_t(a)$ with a proxy loss $\widehat{h}_t(a)$.

The proxy may be obtained from a labeled training split by regression, isotonic calibration, multicalibration, or any other supervised procedure. The certification theorem below does not require $\hat{h}$ to be conservative or well calibrated. Better proxies reduce cost; held-out calibration supplies validity.

Given proxy $\widehat{h}_t(a)$, the implementable rule is the plug-in version of (5):

$$a_t^\lambda \in \arg\min_{a\in\mathcal{A}}\{c_t(a) + \lambda\widehat{h}_t(a)\}.$$

Equivalently, the update rule first chooses the model with the lowest proxy loss,

$$j_t^\lambda \in \arg\min_{j\in[k]} \widehat{h}_t(j), \tag{6}$$

then defers to the expert if and only if $\widehat{h}_t(j_t^\lambda) \geq 1/\lambda$. Denote $D_t^\lambda = \mathbf{1}\{\widehat{h}_t(j_t^\lambda) \geq 1/\lambda\}$; then the action is

$$a_t^\lambda = \begin{cases} E, & D_t^\lambda = 1, \\ j_t^\lambda, & D_t^\lambda = 0. \end{cases} \tag{7}$$

The realized labeling loss and the cost given a price λ are then

$$H_t(\lambda) = (1 - D_t^\lambda)\ell\big(Y_t, \widehat{Y}_t^{(j_t^\lambda)}\big), \qquad C_t(\lambda) = D_t^\lambda. \tag{8}$$

Then the average loss and cost of the price-λ policy are

$$L_T(\lambda) = \frac{1}{T}\sum_{t=1}^{T} H_t(\lambda), \qquad C_T(\lambda) = \frac{1}{T}\sum_{t=1}^{T} C_t(\lambda). \tag{9}$$

2.2 Price Estimation on A Calibration Set

We sample a calibration set $I_1, \dots, I_m$, define the empirical labeling loss

$$\widehat{L}_m(\lambda) = \frac{1}{m}\sum_{s=1}^{m} H_{I_s}(\lambda). \tag{10}$$

Algorithm 1 Offline fixed-price labeling

Require: Unlabeled pool $X_1, \dots, X_T$; cheap predictions $\{\widehat{Y}_t^{(j)}\}_{j\in[k]}$; proxy losses $\{\widehat{h}_t(j)\}_{j\in[k]}$; target error ε; failure level α.

Stage 1: Price estimation on a calibration set

1: Sample calibration indices $I_1, \dots, I_m \sim \mathrm{Unif}([T])$ and collect expert labels $\{Y_{I_s}\}_{s=1}^m$.

2: Form the empirical loss $\widehat{L}_m(\lambda)$ (10) and the level-$(1-\alpha)$ upper confidence bound $U_m(\lambda;\alpha)$ (11).

3: Select the certified price $\hat{\lambda} = \min\{\lambda \geq 0 : U_m(\lambda;\alpha) \leq \varepsilon\}$ (12).

Stage 2: Label the pool at price $\hat{\lambda}$

4: **for** $t = 1, \dots, T$ **do**

5: Pick the best cheap model $j_t \in \arg\min_{j\in[k]} \widehat{h}_t(j)$.

6: **If** $\widehat{h}_t(j_t) \geq 1/\hat{\lambda}$: query expert, $\widetilde{Y}_t \leftarrow Y_t$.

7: **Else**: accept cheap label, $\widetilde{Y}_t \leftarrow \widehat{Y}_t^{(j_t)}$.

8: **end for**

Ensure: Labels $\widetilde{Y}_1, \dots, \widetilde{Y}_T$ with certified error $\leq \varepsilon$ at level $1-\alpha$.

Because $\widehat{L}_m(\lambda)$ is monotone in λ, certifying the entire family costs no more than certifying a single price. We use the Hoeffding bound to give a level-$(1-\alpha)$ upper confidence bound on $L_T(\lambda)$,

$$U_m(\lambda;\alpha) = \widehat{L}_m(\lambda) + \sqrt{\frac{\log(1/\alpha)}{2m}}. \tag{11}$$

Then the price can be estimated by solving the offline LP dual problem with the confidence upper bound $U_m(\lambda;\alpha)$ in place of the true loss $L_T(\lambda)$:

$$\hat{\lambda} = \min\left\{\lambda \geq 0 : U_m(\lambda;\alpha) \leq \varepsilon\right\}, \tag{12}$$

falling back to the expert-only policy $\lambda = \infty$ if no finite price qualifies. Because $C_T(\lambda)$ is non-decreasing and $U_m(\lambda;\alpha)$ non-increasing in λ, (12) coincides with the cost-minimizing selection $\arg\min_\lambda\{C_T(\lambda) : U_m(\lambda;\alpha) \leq \varepsilon\}$.

The overall offline labeling algorithm is summarized in Algorithm 1.

Relation to PAC labeling Under the single-source case, if we use uncertainty score as the proxy, $\widehat{h}_t = U_t$, then AI label is used when

$$\widehat{h}_t = U_t < 1/\hat{\lambda}.$$

This corresponds to the threshold policy used in the original PAC labeling method [1]. Differently, the threshold used in our method is defined by the dual price motivated from the LP perspective. With multiple sources, $\widehat{h}_t$ is replaced by the itemwise minimum of the proxy losses.

2.3 Offline PAC Guarantees

Theorem 1 (Offline PAC guarantee)**.** *Suppose the proxy $\widehat{h}$ is independent of the calibration labels. Let $\hat{\lambda}$ be selected by (12). Then, with probability at least $1-\alpha$ over the draw of the calibration indices, the labeling algorithm produce approximately correct labels*

$$\frac{1}{T}\sum_{t=1}^{T} \ell(Y_t, \widetilde{Y}_t) \leq \varepsilon. \tag{13}$$

The proof is in Appendix B. Validity comes entirely from the held-out calibration labels and is independent of the quality of the proxy; the selected price needs no correction for the search over λ.

Corollary 1 (Generalization guarantee)**.** *Suppose the calibration set is an i.i.d. sample from a distribution P. For a price λ, define the population risk*

$$L(\lambda) = \mathbb{E}_{(X,Y)\sim P}\left[(1 - D^{\lambda}(X))\ell\big(Y, \widehat{Y}^{(j^{\lambda}(X))}(X)\big)\right].$$

Let Λ be a finite price grid fixed independently of the calibration labels, and let $\hat{\lambda}$ be selected over Λ by (12). Then $L(\hat{\lambda}) \leq \varepsilon$ with probability at least $1 - \alpha$.

3 Online PAC Labeling via Audits

The offline guarantee certifies a fixed policy after a labeled calibration stage. In an online data-curation problem, inputs arrive sequentially, decisions are made before future data are seen, and the loss of an AI-labeled item is unobserved unless an expert label is revealed. Therefore the offline calibration argument cannot be reused directly.

The natural online analogue is to keep a moving PAC-labeling threshold. We propose a framework for online PAC labeling in which randomized audits provide the intermittent feedback needed to keep realized error controlled.

3.1 Online PAC Labeling Setup and Audited Feedback

Following the action-cost notation in (2), we focus on the regime

$$c_t(j) = 0 \quad \text{for all } t, j \in [k], \qquad c_t(E) = 1.$$

The online goal is therefore the same as PAC labeling: label as many items as possible with AI while controlling the average error of the released dataset. The difference is that data arrive sequentially, so the uncertainty threshold is updated over time.

At time t, the learner observes the input X_t, cheap model labels $\{\widehat{Y}_t^{(j)}\}_{j=1}^{k}$, and predictable proxy losses $\{\widehat{h}_t(j)\}_{j=1}^{k}$. The proxy losses may be uncertainty scores, calibrated error estimates, or any predictable scores built from past expert/audit labels and current non-expert information. (LT: This is the delicate point: $\widehat{h}_t(j)$ may use X_t, cheap model outputs, past expert labels, and past audits, but it cannot use the current expert label Y_t, the current audit outcome A_t, or the current loss before the routing and audit decisions are made.) The algorithm first routes the item to the model with the smallest proxy loss:

$$j_t^{\lambda} \in \arg\min_{j\in[k]} \widehat{h}_t(j), \qquad \widehat{h}_t^{\lambda} = \widehat{h}_t(j_t^{\lambda}). \tag{14}$$

It then applies a scalar PAC-labeling threshold. Let $\lambda_t \geq 0$ denote the current price of error. Equivalently, $1/\lambda_t$ is the current uncertainty threshold, with the convention that $1/\lambda_t = \infty$ when $\lambda_t = 0$. The deferral decision is

$$D_t = \mathbf{1}\{\lambda_t \widehat{h}_t^{\lambda} \geq 1\} = \mathbf{1}\{\widehat{h}_t^{\lambda} \geq 1/\lambda_t\}. \tag{15}$$

If $D_t = 1$, the item is sent to the expert and the released label is $\widetilde{Y}_t = Y_t$. If $D_t = 0$, the algorithm initially releases the routed AI label $\widetilde{Y}_t = \widehat{Y}_t^{(j_t^\lambda)}$. This is exactly the PAC-labeling threshold form. The multi-model extension is contained in the routed score $\widehat{h}_t^\lambda = \min_j \widehat{h}_t(j)$.

The main online obstacle is partial feedback. When an AI label is used, its true loss is unobserved unless the expert label is also revealed. Therefore, after every AI decision, the algorithm uses an audit probability chosen in advance of the audit draw and before observing the current expert label:

$$A_t \sim \mathrm{Bernoulli}(p_t), \qquad p_t \in [p_{\min}, 1].$$

If $A_t = 1$, the expert label is queried, the final released label is corrected to Y_t, and the audit records the counterfactual error of the routed AI label.

Define the counterfactual AI-decision loss

$$H_t = (1 - D_t)\ell\big(Y_t, \widehat{Y}_t^{(j_t^\lambda)}\big).$$

Deferral has zero loss because the expert label is used. An unaudited AI decision may have nonzero loss, but that loss is not observed. The update uses an internal target $\varepsilon_0 \le \varepsilon$:

$$\widehat{Z}_t = \begin{cases} -\varepsilon_0, & \text{if the item is deferred to the expert,} \\ \{\ell(Y_t, \widehat{Y}_t^{(j_t^\lambda)}) - \varepsilon_0\}/p_t, & \text{if the AI label is used and audited,} \\ 0, & \text{if the AI label is used and not audited.} \end{cases}$$

The role of ε_0 is to leave room for finite-horizon statistical guarantee: the algorithm updates toward the internal target ε_0, while the final PAC target is ε. Corollary 2 gives the condition under which choosing ε_0 yields final error at most ε. The price update is

$$\lambda_{t+1} = [\lambda_t + \eta \widehat{Z}_t]_+, \tag{16}$$

where $\eta > 0$ is a step size. When audited AI mistakes occur more often than the internal target ε_0, the update increases λ_t, lowers the threshold $1/\lambda_t$, and makes future routing more conservative. When deferrals or low-error audits dominate, the price decreases and the algorithm allows more AI labels.

Let $G_t = \ell(Y_t, \widetilde{Y}_t)$ be the realized final loss after audit-and-correct. Since audited AI errors are corrected (so loss becomes zero), and expert deferrals use the true label,

$$G_t \le H_t.$$

The reported final error is the realized error G_t after audit-and-correct, whereas the counterfactual AI error tracks how often the routed AI label would have been wrong before correction. This distinction is important: audits serve both as intermittent feedback for the moving PAC threshold and as corrections for the final dataset.

3.2 Online Finite-Horizon Guarantee

The online result is distribution-free with respect to the data sequence; the probability is over the algorithm's audit randomness. It controls realized error, not regret against an offline expert-query oracle.

Algorithm 2 Audited online PAC labeling

Require: Stream horizon T; target error $\varepsilon \in [0,1]$; audit floor $p_{\min}$; step size η; internal target $\varepsilon_0 \in [0,\varepsilon]$.

Initialize $\lambda_1 \geq 0$.
for $t = 1, \ldots, T$ **do**
  Observe X_t, cheap labels $\{\widehat{Y}_t^{(j)}\}_{j=1}^k$, and predictable proxies $\{\widehat{h}_t(j)\}_{j=1}^k$.
  Choose j_t^λ and $\widehat{h}_t^\lambda$ by (14).
  if $\lambda_t \widehat{h}_t^\lambda \geq 1$ **then**
    Query the expert, output $\widetilde{Y}_t = Y_t$, set $D_t = 1$ and $\widehat{Z}_t = -\varepsilon_0$.
  else
    Output the AI label $\widetilde{Y}_t = \widehat{Y}_t^{(j_t^\lambda)}$ and set $D_t = 0$.
    Choose $p_t \in [p_{\min}, 1]$ before observing Y_t, then draw $A_t \sim \mathrm{Bernoulli}(p_t)$.
    if $A_t = 1$ **then**
      Query the expert, correct $\widetilde{Y}_t = Y_t$, and set $\widehat{Z}_t = \{\ell(Y_t, \widehat{Y}_t^{(j_t^\lambda)}) - \varepsilon_0\}/p_t$.
    else
      Set $\widehat{Z}_t = 0$.
    end if
  end if
  Update λ_{t+1} by (16).
end for

Assumption 1 (Predictable routing and random audits)**.** *The routing decision, audit probability, and selected source are chosen using only past information and the current non-expert information* $(X_t, \{\widehat{Y}_t^{(j)}, \widehat{h}_t(j)\}_{j=1}^k)$*. After this decision, the audit draw is independent Bernoulli with known probability* $p_t \in [p_{\min}, 1]$*.*

Assumption 2 (Proxy floor)**.** *For some* $\rho > 0$*,* $\widehat{h}_t(j) \in [\rho, 1]$ *for all* t, j*.*

For the finite-horizon analysis, define the augmented observation mask and its conditional observation probability by

$$O_t = D_t + (1 - D_t)A_t, \qquad q_t = D_t + (1 - D_t)p_t. \tag{17}$$

Thus $q_t = 1$ on expert deferrals and $q_t = p_t$ on AI decisions. The three-case update in Algorithm 2 can equivalently be written as the centered inverse-observation signal

$$\widehat{Z}_t = \frac{O_t}{q_t}(H_t - \varepsilon_0). \tag{18}$$

Let $\mathcal{G}_t$ denote the pre-audit sigma-field that contains the history through round $t-1$, the current non-expert information, the selected source j_t^λ, the deferral decision D_t, the audit probability p_t, and the latent selected loss. It does not contain the current audit draw A_t. The audit assumption says that, conditional on $\mathcal{G}_t$, A_t is an independent Bernoulli random variable with mean p_t whenever $D_t = 0$.

Proposition 1 (Unbiased intermittent update signal)**.** *Under bounded loss and online predictability,*

$$\mathbb{E}\left[\frac{O_t}{q_t} \,\middle|\, \mathcal{G}_t\right] = 1, \qquad \mathbb{E}[\widehat{Z}_t \mid \mathcal{G}_t] = H_t - \varepsilon_0.$$

Theorem 2 (Audited online finite-horizon control)**.** *Assume bounded loss, online predictability and audits, and the proxy floor. Let*

$$\lambda_{\mathrm{def}} = \frac{1}{\rho}, \qquad \Lambda_{\mathrm{on}} = \max\left\{\lambda_1, \lambda_{\mathrm{def}} + \frac{\eta(1-\varepsilon_0)}{p_{\min}}\right\}.$$

Run Algorithm 2 with constant step size $\eta > 0$ and internal target $\varepsilon_0 \in [0, \varepsilon]$, where $\varepsilon \in [0, 1]$. Then, for any fixed horizon T and any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the audit randomness,

$$\frac{1}{T}\sum_{t=1}^{T} H_t \le \varepsilon_0 + \frac{\Lambda_{\mathrm{on}}}{\eta T} + \frac{1}{p_{\min}}\sqrt{\frac{2\log(1/\delta)}{T}}. \tag{19}$$

Consequently, if audited AI labels are corrected to expert labels in the final dataset, then

$$\frac{1}{T}\sum_{t=1}^{T} G_t \le \varepsilon_0 + \frac{\Lambda_{\mathrm{on}}}{\eta T} + \frac{1}{p_{\min}}\sqrt{\frac{2\log(1/\delta)}{T}}. \tag{20}$$

Corollary 2 (PAC-style online tuning)**.** *Under the conditions of Theorem 2, suppose*

$$\varepsilon_0 + \frac{\Lambda_{\mathrm{on}}}{\eta T} + \frac{1}{p_{\min}}\sqrt{\frac{2\log(1/\delta)}{T}} \le \varepsilon,$$

with $\varepsilon_0 \ge 0$. Then the audit-and-correct final labels satisfy

$$\frac{1}{T}\sum_{t=1}^{T} G_t \le \varepsilon$$

with probability at least $1 - \delta$. With $\eta \asymp T^{-1/2}$ and constant $p_{\min}$, the slack is $O(T^{-1/2})$.

The proofs of Proposition 1, Theorem 2, and Corollary 2 are in Appendix B.2.

3.3 Offline versus Online Certification

The online theorem is deliberately different from the offline one. Offline calibration certifies a fixed candidate policy by evaluating it on a held-out labeled sample. Online auditing controls the long-run error of a moving policy by converting unobserved AI losses into an unbiased intermittent-feedback signal. The proxy losses again affect efficiency rather than validity, but in the online setting some audit or reveal mechanism is essential: without it, two streams can induce identical observations while having opposite AI-label correctness on every unaudited item.

4 Experiments

We run a small proof-of-concept study on ImageNet and ImageNetV2 using the confidence score already included in the data. There is one cheap source, namely the model prediction, and one expert source, namely the ground-truth label. The cheap-source proxy loss is $\widehat{h} = 1 - \text{confidence}$, the cheap source has zero marginal cost, and the expert has unit cost. All reported numbers average over 500 random trials. Appendix C gives the full sweep and implementation details.

Offline calibration. The offline experiment compares the fixed-price rule from Section 2 with the original scalar PAC threshold baseline. Because this experiment has only one cheap source, the fixed-price policies are nested uncertainty-threshold rules. We therefore certify the fixed-price rule with the sharper monotone confidence bound

$$\hat{L}_m(\lambda) + \sqrt{\frac{\log(1/\alpha)}{2m}},$$

rather than the finite-class union bound used for general multi-source policy classes. At $\varepsilon = 0.10$, both methods have zero violations in 500 trials and nearly identical expert-budget savings (Table 1). Figure 1 shows the same comparison across all three ε levels, and Table 4 gives the corresponding full sweep.

Dataset	Method	Error	Violation	Budget saved
ImageNet	fixed-price monotone UCB	0.0873	0.0000	0.7848
ImageNet	original PAC	0.0879	0.0000	0.7864
ImageNetV2	fixed-price monotone UCB	0.0718	0.0000	0.5457
ImageNetV2	original PAC	0.0726	0.0000	0.5479

Table 1: Offline results at $\varepsilon = 0.10$. "Budget saved" is one minus the fraction of items sent to the expert.

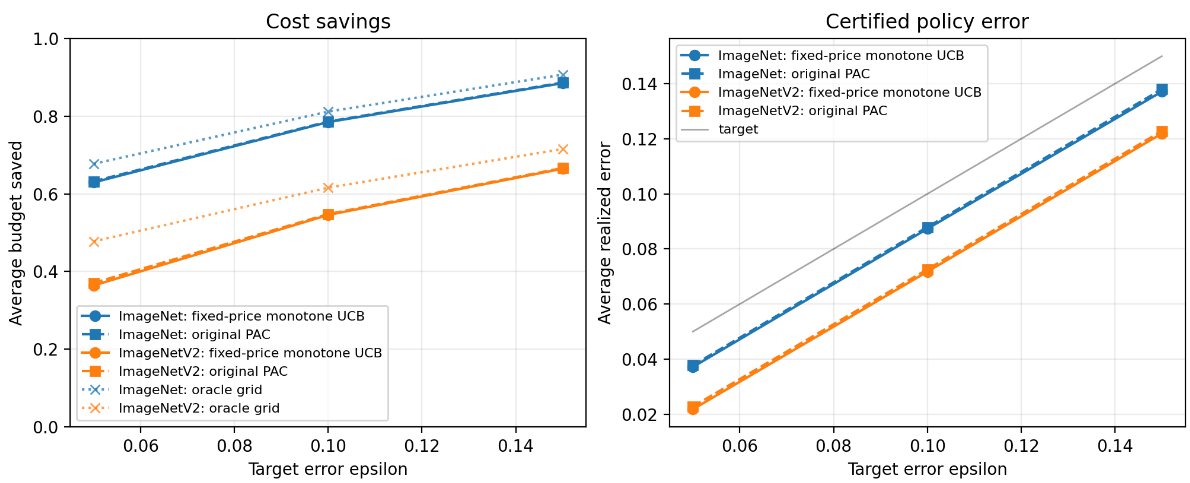

Figure 1: Offline sweep over target error levels. After exploiting the single-source monotonicity, the fixed-price rule is nearly indistinguishable from the original scalar PAC baseline; the remaining gap comes from using a fixed threshold grid rather than the exact calibration-order threshold.

Two-source synthetic routing. The single-source experiment above is the special case in which the LP and the original scalar PAC threshold coincide. To stress-test the multi-source regime that motivates the LP formulation, we synthesize two complementary cheap sources from each base ImageNet table. Items are partitioned by the parity of the true class label. On items inside its specialty each source reuses the original predictor's label and confidence; on items outside its specialty each source emits a uniform-random label and a confidence drawn from Uniform$[0, 0.2]$, so the source is calibrated to its own competence. Source A's specialty is the even-class half and source B's is the odd-class half. Neither source dominates; standalone accuracy is approximately 0.39 on

ImageNet and 0.32 on ImageNetV2, while the per-item oracle that picks whichever source is correct attains 0.78 and 0.65, respectively. Costs match the simple offline experiment: $c_A = c_B = 0$, $c_E = 1$, so the dual rule routes to $\arg\min_j \widehat{h}_t(j)$ and defers when $\min_j \widehat{h}_t(j) > 1/\lambda$. We compare three calibrated policies. *LP two-source uniform UCB* is the fixed-price rule of Section 2.2 applied to the constructed two-source pool, certified with the conservative finite-class Hoeffding bound (25) over a $|\Lambda| = 202$ threshold grid; we report it as a worst-case baseline. We emphasize, however, that with $c_A = c_B = 0$ the routed loss *is* monotone in the scalar score $\widehat{h}_t^\lambda = \min_j \widehat{h}_t(j)$, because the source map $\arg\min_j \widehat{h}_t(j)$ does not depend on λ, so Theorem 1 licenses the sharper single-price bound (11) here as well; this replaces the $\sqrt{\log(|\Lambda|/\alpha)}$ radius by $\sqrt{\log(1/\alpha)}$ and is most consequential in the tight-ε regime discussed below. *Source-j monotone PAC* ($j \in \{A, B\}$) is the original PAC baseline restricted to one source. *Best-of-singles (Bonferroni)* runs both single-source baselines at level $\alpha/2$ and picks the source with the larger empirical savings on the calibration sample, which remains a valid α-PAC procedure.

Dataset	Method	Error	Violation	Budget saved
ImageNet	LP two-source uniform UCB	0.0793	0.0000	**0.7645**
ImageNet	Best-of-singles (Bonferroni)	0.0879	0.0000	0.4746
ImageNet	Source A monotone PAC	0.0878	0.0000	0.4715
ImageNet	Source B monotone PAC	0.0877	0.0000	0.4754
ImageNetV2	LP two-source uniform UCB	0.0534	0.0000	**0.4923**
ImageNetV2	Best-of-singles (Bonferroni)	0.0720	0.0000	0.3587
ImageNetV2	Source A monotone PAC	0.0727	0.0000	0.3470
ImageNetV2	Source B monotone PAC	0.0730	0.0000	0.3629

Table 2: Two-source synthetic routing at $\varepsilon = 0.10$. The LP method routes each item to the source with the lower proxy loss and certifies the routed average error against the union bound over the 202-threshold grid. The largest budget saved in each dataset is shown in bold.

At $\varepsilon = 0.10$ all four methods have zero violations in 500 trials. The LP rule saves roughly 0.29 more of the expert budget on ImageNet and 0.13 more on ImageNetV2 than the best Bonferroni-corrected single-source baseline (Table 2). The gap on ImageNet is close to the LP oracle's envelope (0.81 saved at $\varepsilon = 0.10$), confirming that the per-item routing benefit dominates the finite-policy union-bound radius once the calibration sample is large enough. At $\varepsilon = 0.05$ on ImageNetV2 the order reverses: with $m = 2000$ the uniform radius $\sqrt{\log(202/\alpha)/(2m)} \approx 0.046$ leaves almost no headroom under $\varepsilon = 0.05$, so the LP rule certifies near-zero savings while each single-source baseline still certifies its specialty half (Figure 2; Table 5 in the appendix reports the full sweep). With equal cheap-source costs this gap is removed rather than merely shrunk: the routed loss is monotone in the unidimensional statistic $\widehat{h}_t^\lambda = \min_j \widehat{h}_t(j)$, so Theorem 1 certifies with the single-price radius $\sqrt{\log(1/\alpha)/(2m)}$ and pays no $\sqrt{\log|\Lambda|}$ penalty. The uniform-UCB numbers above are therefore a conservative baseline; the monotone certificate recovers the headroom lost at small ε. (A variance-adaptive Bernstein bound would tighten the constant further; we leave that to future work.)

Online audited routing. The online experiment uses a random stream order and the dual-price update in Section 3 with audit probabilities $p \in \{0.1, 0.2, 0.5\}$. The update follows the intermittent-feedback form: an unaudited AI decision does not move the price, while an expert deferral is treated as an observed zero-loss event. These runs set the internal online target to $\varepsilon_0 = \varepsilon$, so they are a behavioral check of the audited update rather than a theorem-tight slack-adjusted implementation

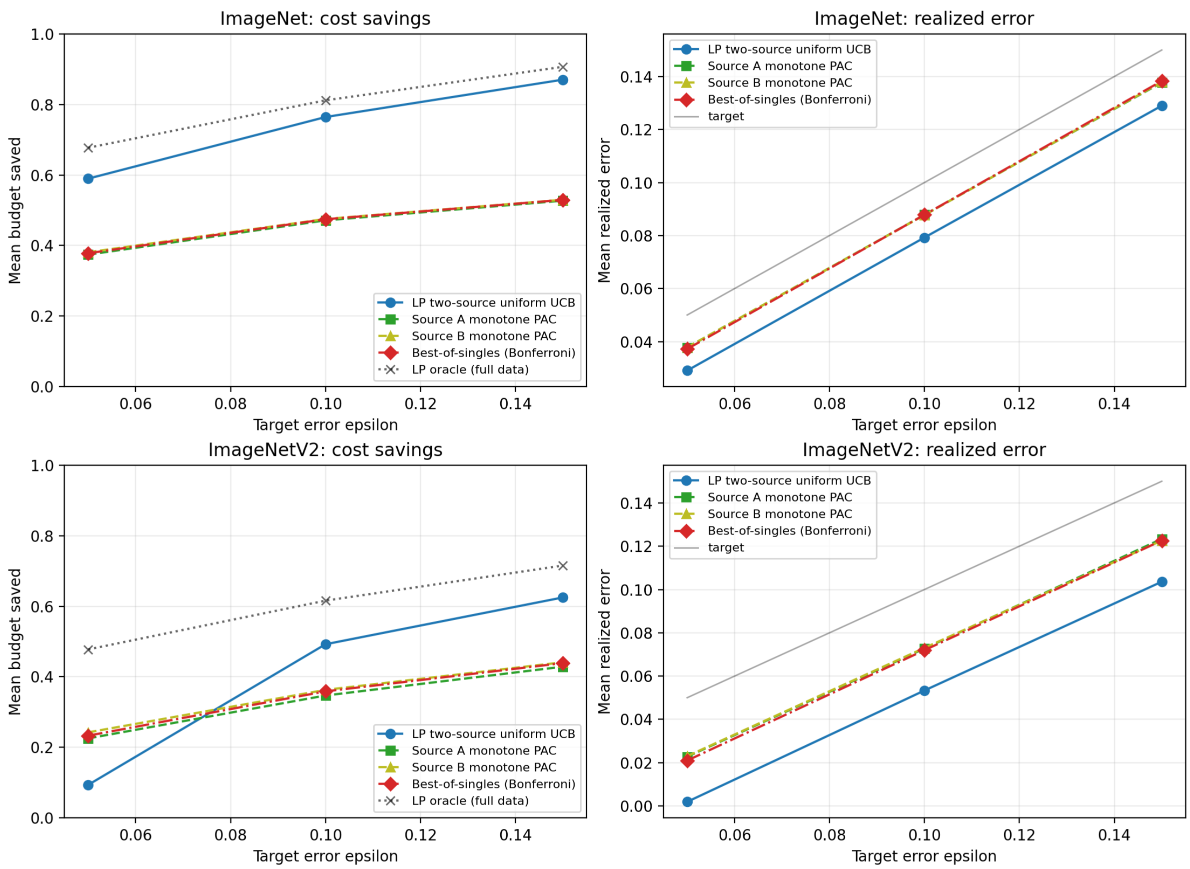

Figure 2: Two-source synthetic routing sweep. Each single source can certify at most its own specialty half, so the three single-source curves saturate near 0.55 (ImageNet) and 0.45 (ImageNetV2). The LP method recovers most of the per-item oracle envelope by routing across sources at the cost of a $\sqrt{\log|\Lambda|}$ uniform-certification radius. On ImageNetV2 at $\varepsilon = 0.05$ that radius approaches ε itself and the LP method is forced into near-full deferral, illustrating where a tighter sup-deviation bound would be needed.

of Corollary 2. Lower audit rates save more expert budget but leave less correction, so final error is closer to the target. At $\varepsilon = 0.10$, ImageNet has no final-error violations for all three audit rates; ImageNetV2 has a small violation rate at $p = 0.1$ and none at $p \geq 0.2$ (Table 3; the full sweep across $\varepsilon \in \{0.05, 0.10, 0.15\}$ is in Table 6).

Dataset	p	Final error	AI error	Violation	Budget saved
ImageNet	0.1	0.0780	0.0866	0.0000	0.6629
ImageNet	0.2	0.0737	0.0921	0.0000	0.6115
ImageNet	0.5	0.0490	0.0981	0.0000	0.3952
ImageNetV2	0.1	0.0863	0.0958	0.0400	0.5091
ImageNetV2	0.2	0.0783	0.0979	0.0000	0.4681
ImageNetV2	0.5	0.0499	0.0996	0.0000	0.3010

Table 3: Online audited results at $\varepsilon = 0.10$. "AI error" is the counterfactual average error of AI-labeled items before audit correction; "Final error" is after audited items are corrected by the expert.

5 Discussion

The report separates two regimes. The offline method has a clean division of labor. The proxy model and the LP structure make the policy cost efficient; the calibration set makes it valid. When instantiating the method, the main modeling choice is whether the target is transductive or inductive. If the application is dataset curation for a fixed unlabeled pool, the transductive theorem should be the main result. If the application is repeated deployment on future items, the inductive statement should be foregrounded. The online method addresses a different obstacle: the selected policy changes over time and AI losses are not observed without expert reveal. Random audits supply the missing feedback through an inverse-probability loss estimate. This gives finite-horizon error control, but not a cost-regret comparison to the offline oracle LP. The experiments in Section 4 illustrate this distinction: offline calibration gives a clean finite-policy certificate, while the online mechanism exposes an explicit audit-cost versus correction tradeoff. The main empirical limitation is that the current data contain only one cheap source. A stronger test of the LP perspective would use several cheap sources with heterogeneous costs, where the original scalar threshold structure no longer captures the routing problem.

A Related Work

PAC labels and conformal calibration. The closest starting point is PAC labeling [1], which gives finite-sample guarantees for cost-effective dataset labeling using AI predictions and expert audits. That framework already includes both scalar-threshold PAC labeling and multi-model PAC routing variants. Our formulation is best viewed as a complementary optimization-first presentation: we start from a finite-pool cost-minimization LP, derive the fixed-price routing rule from its dual, allow item- and source-specific costs, and then use calibration only to certify the actual loss of the selected proxy-generated policy. The calibration logic is closely related to conformal prediction and distribution-free uncertainty quantification [2, 3], but the object being controlled here is the average error of a released labeled dataset rather than coverage of prediction sets.

Online conformal prediction with partial feedback. Adaptive and online conformal methods update calibration parameters over time to handle changing environments [4]. Recent work has studied intermittent or semi-bandit feedback, where labels are not always observed [5, 6, 7]. Those papers motivate the audited online extension in Section 3: when labels are not always revealed, the feedback used to update a calibration or price parameter must be corrected for the observation process.

Learning to defer and cost-sensitive routing. Learning-to-defer methods train systems that either predict or pass an item to an expert [8, 9]. Multiple-expert and cost-sensitive variants study richer deferral and routing settings [10, 11]. The distinction here is the finite-sample PAC certificate for the final released labels. The proxy router may be learned by standard supervised methods, but the guarantee comes from held-out calibration of the actual selected policy.

B Proofs

B.1 Dual Derivation

Introduce one dual variable $\lambda \geq 0$ for the average-error constraint and one free dual variable $\mu_t \in \mathbb{R}$ for each simplex equality. The nonnegativity constraints $x_{t,a} \geq 0$ are kept as the domain of the inner minimization.

Primal constraint	Dual variable	Sign restriction
$\frac{1}{T}\sum_{t,a} h_t(a)x_{t,a} \leq \varepsilon$	λ	$\lambda \geq 0$
$\sum_a x_{t,a} = 1$ for item t	μ_t	$\mu_t \in \mathbb{R}$
$x_{t,a} \geq 0$	domain constraint	–

Using the equality convention $\mu_t(1 - \sum_a x_{t,a})$, the Lagrangian is

$$\begin{aligned}\mathcal{L}(x;\lambda,\mu) =& \frac{1}{T}\sum_{t,a} c_t(a)x_{t,a} + \lambda\left(\frac{1}{T}\sum_{t,a} h_t(a)x_{t,a} - \varepsilon\right) + \sum_{t=1}^{T}\mu_t\left(1 - \sum_a x_{t,a}\right)\\ =& \sum_{t,a}\left[\frac{1}{T}\{c_t(a) + \lambda h_t(a)\} - \mu_t\right]x_{t,a} + \sum_{t=1}^{T}\mu_t - \lambda\varepsilon. \end{aligned} \tag{21}$$

For fixed (λ, μ), the inner problem is $\inf_{x\geq 0}\mathcal{L}(x;\lambda,\mu)$. Since the Lagrangian is linear in every $x_{t,a}$, this infimum is finite if and only if

$$\mu_t \leq \frac{1}{T}\{c_t(a) + \lambda h_t(a)\}, \qquad t \in [T],\ a \in \mathcal{A}. \tag{22}$$

Otherwise some variable with a negative coefficient can be sent to $+\infty$, driving the Lagrangian to $-\infty$. When (22) holds, the x terms vanish at the inner minimum and the dual function is

$$g(\lambda,\mu) = \sum_{t=1}^{T}\mu_t - \lambda\varepsilon.$$

Thus the explicit dual LP is

$$\begin{aligned}
\text{maximize}_{\lambda,\mu} \quad & \sum_{t=1}^{T} \mu_t - \lambda\varepsilon \\
\text{subject to} \quad & \mu_t \le \frac{1}{T}\{c_t(a) + \lambda h_t(a)\}, \qquad t \in [T],\ a \in \mathcal{A}, \\
& \lambda \ge 0, \qquad \mu_t \in \mathbb{R},\ t \in [T].
\end{aligned} \tag{23}$$

For any fixed λ, the objective is increasing in each μ_t, so the optimal value is attained at

$$\mu_t^\star(\lambda) = \frac{1}{T}\min_{a\in\mathcal{A}}\{c_t(a) + \lambda h_t(a)\}.$$

Substitution gives the reduced dual (4). The primal is feasible because choosing the expert for all items gives zero loss, and it is bounded because costs are nonnegative and finite. Therefore strong duality holds by finite-dimensional linear programming duality.

Proof of Theorem 1. Condition on the target pool and the proxy; since $\widehat{h}$ does not use the calibration labels, the routed source $j_t^\lambda \in \arg\min_j \widehat{h}_t(j)$, the routed score $s_t := \widehat{h}_t(j_t^\lambda) = \min_j \widehat{h}_t(j)$, and the routed losses $\ell(Y_t, \widehat{Y}_t^{(j_t^\lambda)}) \in [0,1]$ are all deterministic. Under equal cheap-source costs the routed source does not depend on λ, so the price enters only through the deferral rule $D_t^\lambda = \mathbf{1}\{s_t \ge 1/\lambda\}$.

Monotonicity. As λ increases, $1/\lambda$ decreases, so the deferred set $\{t : s_t \ge 1/\lambda\}$ grows and the routed set $\{t : s_t < 1/\lambda\}$ shrinks: the induced policies form a nested family. A deferred item contributes zero loss and a routed item contributes its fixed loss, so each $H_t(\lambda) = (1 - D_t^\lambda)\ell(Y_t, \widehat{Y}_t^{(j_t^\lambda)})$ is non-increasing in λ, and hence so are $L_T(\lambda)$ and $\widehat{L}_m(\lambda)$. Both change only at the finitely many breakpoints $\{1/s_t\}$, so it suffices to consider λ over this finite set.

Per-price validity. For each fixed λ the calibration draws $H_{I_s}(\lambda)$ are i.i.d. in $[0,1]$ with mean $L_T(\lambda)$, so Hoeffding's inequality gives the single-price bound, valid one λ at a time,

$$\mathbb{P}\big\{L_T(\lambda) > U_m(\lambda;\alpha)\big\} \le \alpha, \qquad U_m(\lambda;\alpha) = \widehat{L}_m(\lambda) + \sqrt{\tfrac{\log(1/\alpha)}{2m}}.$$

Single-point reduction. Let λ° be the largest breakpoint with $L_T(\lambda^\circ) > \varepsilon$; if no such price exists then even $\lambda = 0$ is feasible and the claim is immediate. By monotonicity every price above λ° is feasible, so $L_T(\hat\lambda) > \varepsilon$ can hold only if $\hat\lambda \le \lambda^\circ$. On that event, since $\hat\lambda$ is the smallest price with $U_m(\cdot;\alpha) \le \varepsilon$ and $U_m(\cdot;\alpha)$ is non-increasing, $U_m(\lambda^\circ;\alpha) \le U_m(\hat\lambda;\alpha) \le \varepsilon$; but $L_T(\lambda^\circ) > \varepsilon$ by definition. Hence the bound fails at the single fixed price λ°, an event of probability at most α by single-price validity. Therefore $\mathbb{P}\{L_T(\hat\lambda) \le \varepsilon\} \ge 1 - \alpha$.

Finally, the released label equals $\widehat{Y}_t^{(j_t^{\hat\lambda})}$ on routed items and Y_t (zero loss) on deferred items, so $\frac{1}{T}\sum_t \ell(Y_t, \widetilde{Y}_t) = L_T(\hat\lambda)$, and correcting calibration items to their expert labels can only lower this. Integrating over the proxy preserves the bound. $\square$

Proof of Corollary 1. Fix λ. Under equal cheap-source costs the routed source $j^\lambda(X)$ does not depend on λ and $D^\lambda(X) = \mathbf{1}\{\min_j \widehat{h}(X,j) \ge 1/\lambda\}$ is non-decreasing in λ, so the integrand

$(1 - D^\lambda(X))\,\ell(Y, \widehat{Y}^{(j^\lambda(X))}(X)) \in [0,1]$ is non-increasing in λ pointwise; hence $L(\lambda)$ is non-increasing. For each fixed λ the calibration losses $H_{I_s}(\lambda)$ are i.i.d. in $[0,1]$ with mean $L(\lambda)$, so the single-price bound (11) is a valid upper bound on $L(\lambda)$, one λ at a time. Because Λ is finite, $\lambda^\circ = \max\{\lambda \in \Lambda : L(\lambda) > \varepsilon\}$ is attained (or no such price exists, in which case the claim is immediate), and the single-point argument of Theorem 1—now applied to $L(\lambda)$ on Λ—gives $L(\hat{\lambda}) \le \varepsilon$ with probability at least $1-\alpha$ over the calibration sample.

Condition on the calibration sample and on the event $L(\hat{\lambda}) \le \varepsilon$. The deployment losses $\ell(Y_t, \widetilde{Y}_t)$ are then independent, bounded in $[0,1]$, and have conditional mean at most $L(\hat{\lambda}) \le \varepsilon$, so Hoeffding's inequality gives

$$\frac{1}{T}\sum_{t=1}^{T} \ell(Y_t, \widetilde{Y}_t) \le \varepsilon + \sqrt{\frac{\log(1/\delta)}{2T}}$$

with conditional probability at least $1-\delta$. Combining the calibration and deployment events gives probability at least $1-\alpha-\delta$. ☐

B.2 Online Proofs

Proof of Proposition 1. Recall that $\mathcal{G}_t$ fixes the predictable round-t information, the selected source j_t^λ, the routed proxy loss $\widehat{h}_t^\lambda = \widehat{h}_t(j_t^\lambda)$, the audit probability p_t, the deferral decision D_t, and the latent selected loss, but not the audit draw A_t. If $D_t = 1$, then $O_t = q_t = 1$ and $H_t = 0$, so

$$\mathbb{E}\left[\frac{O_t}{q_t} \,\middle|\, \mathcal{G}_t\right] = 1, \qquad \mathbb{E}[\widehat{Z}_t \mid \mathcal{G}_t] = -\varepsilon_0 = H_t - \varepsilon_0.$$

If $D_t = 0$, then $O_t = A_t$, $q_t = p_t$, and conditional on $\mathcal{G}_t$ the audit draw is Bernoulli(p_t) while the loss $\ell(Y_t, \widehat{Y}_t^{(j_t^\lambda)})$ is fixed. Hence

$$\mathbb{E}\left[\frac{O_t}{q_t} \,\middle|\, \mathcal{G}_t\right] = \frac{p_t}{p_t} = 1.$$

Since H_t and ε_0 are fixed conditional on $\mathcal{G}_t$, this also gives

$$\mathbb{E}[\widehat{Z}_t \mid \mathcal{G}_t] = \mathbb{E}\left[\frac{O_t}{q_t}(H_t - \varepsilon_0) \,\middle|\, \mathcal{G}_t\right] = H_t - \varepsilon_0.$$

☐

Proof of Theorem 2. First bound the price process. If $\lambda_t \ge \lambda_{\text{def}} = 1/\rho$, then

$$\lambda_t \widehat{h}_t^\lambda = \lambda_t \widehat{h}_t(j_t^\lambda) \ge \lambda_{\text{def}}\rho = 1.$$

Thus the algorithm defers to the expert, so $D_t = 1$ and $\widehat{Z}_t = -\varepsilon_0$. Consequently $\lambda_{t+1} = [\lambda_t - \eta\varepsilon_0]_+ \le \lambda_t$. If instead $\lambda_t < \lambda_{\text{def}}$, then $\widehat{Z}_t \le (1-\varepsilon_0)/p_{\min}$ because $\varepsilon_0 \in [0,1]$ and the loss is bounded by one. Hence

$$\lambda_{t+1} \le \lambda_t + \eta(1-\varepsilon_0)/p_{\min} < \lambda_{\text{def}} + \eta(1-\varepsilon_0)/p_{\min}.$$

By induction, $0 \le \lambda_t \le \Lambda_{\text{on}}$ for all t.

Since $[x]_+ \ge x$, the update gives

$$\lambda_{t+1} \ge \lambda_t + \eta\widehat{Z}_t.$$

Summing over $t = 1, \ldots, T$ yields

$$\sum_{t=1}^{T} \widehat{Z}_t \leq \frac{\lambda_{T+1} - \lambda_1}{\eta} \leq \frac{\Lambda_{\mathrm{on}}}{\eta}.$$

Let $M_t = (H_t - \varepsilon_0) - \widehat{Z}_t$. By Proposition 1 and iterated expectation, M_t is a martingale difference with respect to the filtration generated by the past rounds and the current audit draw. Moreover, $|M_t| \leq 1/p_{\min}$: if $D_t = 1$, then $M_t = 0$; if $D_t = 0$ and $A_t = 0$, then $|M_t| = |H_t - \varepsilon_0| \leq 1$; and if $D_t = 0$ and $A_t = 1$, then

$$|M_t| = |H_t - \varepsilon_0| \left| \frac{1}{p_t} - 1 \right| \leq \frac{1}{p_t} \leq \frac{1}{p_{\min}}.$$

Azuma-Hoeffding gives, with probability at least $1 - \delta$,

$$\sum_{t=1}^{T} M_t \leq \frac{1}{p_{\min}} \sqrt{2T \log(1/\delta)}.$$

Because

$$\sum_{t=1}^{T} H_t = T\varepsilon_0 + \sum_{t=1}^{T} \widehat{Z}_t + \sum_{t=1}^{T} M_t,$$

combining the last displays gives (19). Finally, audit-and-correct gives $G_t \leq H_t$ pointwise, which proves (20). $\square$

Proof of Corollary 2. The stated condition makes the right-hand side of (20) at most ε. If $\eta \asymp T^{-1/2}$, $p_{\min}$ is constant, and the proxy floor is fixed so that $\Lambda_{\mathrm{on}} = O(1)$, then both excess terms in (20) are $O(T^{-1/2})$. $\square$

C Experimental Details

The experiments use the two ImageNet CSV files from the HW4 copy directory: ImageNet has $T = 50000$ items and ImageNetV2 has $T = 10000$ items. Each file contains a ground-truth label, a model prediction, and a confidence score. We treat the model prediction as the only cheap source and the ground-truth label as the expert. The proxy loss is $\widehat{h} = 1 -$ confidence, the cheap-source deployment cost is zero, and the expert cost is one. All experiments use 500 random trials. The scripts are `run_simple_offline_pac_experiment.py`, `run_simple_online_pac_experiment.py`, and `run_two_source_offline_pac_experiment.py`.

Offline implementation. For the fixed-price method, the candidate threshold grid contains the expert-only rule plus 201 evenly spaced confidence-loss thresholds between 0 and 1. Although Section 2.2 states the general finite-policy bound, this single-source experiment uses the sharper monotone threshold radius

$$\sqrt{\frac{\log(1/\alpha)}{2m}},$$

which is valid uniformly over all scalar thresholds by the same nested-class argument used in the original PAC baseline. For each trial, the calibration indices are sampled independently and uniformly from the target pool with replacement, with calibration size $m = 0.2T$. The confidence level is $\alpha = 0.05$ and $\varepsilon \in \{0.05, 0.10, 0.15\}$. The original PAC baseline uses the same calibration sample and the same monotone radius, but chooses its threshold directly from the calibration mistake order instead of from the fixed grid. The full sweep over $\varepsilon \in \{0.05, 0.10, 0.15\}$ is reported in Table 4.

Dataset	Method	ε	Error	Violation	Budget saved
ImageNet	fixed-price monotone UCB	0.05	0.0372	0.0000	0.6296
ImageNet	original PAC	0.05	0.0377	0.0000	0.6323
ImageNet	fixed-price monotone UCB	0.10	0.0873	0.0000	0.7848
ImageNet	original PAC	0.10	0.0879	0.0000	0.7864
ImageNet	fixed-price monotone UCB	0.15	0.1372	0.0000	0.8851
ImageNet	original PAC	0.15	0.1380	0.0000	0.8866
ImageNetV2	fixed-price monotone UCB	0.05	0.0218	0.0000	0.3642
ImageNetV2	original PAC	0.05	0.0228	0.0000	0.3706
ImageNetV2	fixed-price monotone UCB	0.10	0.0718	0.0000	0.5457
ImageNetV2	original PAC	0.10	0.0726	0.0000	0.5479
ImageNetV2	fixed-price monotone UCB	0.15	0.1219	0.0000	0.6648
ImageNetV2	original PAC	0.15	0.1228	0.0000	0.6665

Table 4: Full offline sweep. Error, violation, and budget saved are averages over 500 trials. Rows are grouped by (dataset, ε) scenario and separated by light rules.

Two-source implementation. `run_two_source_offline_pac_experiment.py` constructs two complementary cheap sources from each base ImageNet table using the parity-of-true-class partition described in Section 4. The construction is seeded separately from the trial sampler so the synthesized pool is fixed across trials; defaults are `--synth-seed 20260526` and `--low-conf-max 0.2`. The calibration grid contains the expert-only policy plus 201 evenly spaced thresholds on $\min_j \widehat{h}_t(j)$, the calibration size is $m = 0.2T$, and $\alpha = 0.05$. The LP method uses the union-bound radius $\sqrt{\log(|\Lambda|/\alpha)/(2m)}$; the single-source baselines use the monotone radius $\sqrt{\log(1/\alpha)/(2m)}$; the Bonferroni-corrected best-of-singles baseline runs both single-source baselines at $\alpha/2$ each and reports the one with larger empirical savings on the calibration sample. Constructed-pool statistics are written to `results/two_source_offline_pac/two_source_offline_pac_construction.csv`. The full sweep over $\varepsilon \in \{0.05, 0.10, 0.15\}$ for all four methods is in Table 5.

Online implementation. For each trial, the target pool is randomly permuted into a stream. The update uses $\eta = 1$, proxy floor $\rho = 0.01$, and $\lambda_1 = 1/\varepsilon$. The online routing rule uses the AI prediction when $\lambda_t \max(1 - \text{confidence}_t, \rho) < 1$ and otherwise defers to the expert. If the AI prediction is used, it is audited with probability $p \in \{0.1, 0.2, 0.5\}$. The hybrid intermittent update is $-\eta\varepsilon_0$ on expert deferrals, $\eta(\ell_t - \varepsilon_0)/p$ on audited AI decisions, and zero on unaudited AI decisions. Audited AI labels are corrected by the expert before release; unaudited AI labels are released as-is. The reported "AI error" is the counterfactual average loss before audit correction, and the reported final error is the realized loss after correction. As noted in Section 4, these runs set $\varepsilon_0 = \varepsilon$ rather than subtracting the finite-horizon slack from Corollary 2. The full sweep over $\varepsilon \in \{0.05, 0.10, 0.15\}$ and $p \in \{0.1, 0.2, 0.5\}$ is reported in Table 6.

Dataset	Method	ε	Error	Violation	Budget saved
ImageNet	LP two-source uniform UCB	0.05	0.0290	0.0000	**0.5898**
ImageNet	Best-of-singles (Bonferroni)	0.05	0.0372	0.0000	0.3782
ImageNet	Source A monotone PAC	0.05	0.0378	0.0000	0.3743
ImageNet	Source B monotone PAC	0.05	0.0378	0.0000	0.3809
ImageNet	LP two-source uniform UCB	0.10	0.0793	0.0000	**0.7645**
ImageNet	Best-of-singles (Bonferroni)	0.10	0.0879	0.0000	0.4746
ImageNet	Source A monotone PAC	0.10	0.0878	0.0000	0.4715
ImageNet	Source B monotone PAC	0.10	0.0877	0.0000	0.4754
ImageNet	LP two-source uniform UCB	0.15	0.1290	0.0000	**0.8705**
ImageNet	Best-of-singles (Bonferroni)	0.15	0.1383	0.0000	0.5296
ImageNet	Source A monotone PAC	0.15	0.1377	0.0000	0.5273
ImageNet	Source B monotone PAC	0.15	0.1379	0.0000	0.5299
ImageNetV2	LP two-source uniform UCB	0.05	0.0019	0.0000	0.0936
ImageNetV2	Best-of-singles (Bonferroni)	0.05	0.0210	0.0000	0.2331
ImageNetV2	Source A monotone PAC	0.05	0.0227	0.0000	0.2254
ImageNetV2	Source B monotone PAC	0.05	0.0230	0.0000	**0.2424**
ImageNetV2	LP two-source uniform UCB	0.10	0.0534	0.0000	**0.4923**
ImageNetV2	Best-of-singles (Bonferroni)	0.10	0.0720	0.0000	0.3587
ImageNetV2	Source A monotone PAC	0.10	0.0727	0.0000	0.3470
ImageNetV2	Source B monotone PAC	0.10	0.0730	0.0000	0.3629
ImageNetV2	LP two-source uniform UCB	0.15	0.1036	0.0000	**0.6250**
ImageNetV2	Best-of-singles (Bonferroni)	0.15	0.1226	0.0000	0.4389
ImageNetV2	Source A monotone PAC	0.15	0.1234	0.0000	0.4287
ImageNetV2	Source B monotone PAC	0.15	0.1226	0.0000	0.4412

Table 5: Full two-source sweep. Error, violation, and budget saved are averages over 500 trials of calibration sampling. Rows are grouped by (dataset, ε) scenario and separated by light rules; the largest budget saved in each scenario is shown in bold.

Dataset	ε	p	Final error	AI error	Violation	Budget saved
ImageNet	0.05	0.1	0.0436	0.0485	0.0100	0.5738
ImageNet	0.05	0.2	0.0394	0.0493	0.0000	0.5244
ImageNet	0.05	0.5	0.0249	0.0497	0.0000	0.3344
ImageNet	0.10	0.1	0.0780	0.0866	0.0000	0.6629
ImageNet	0.10	0.2	0.0737	0.0921	0.0000	0.6115
ImageNet	0.10	0.5	0.0490	0.0981	0.0000	0.3952
ImageNet	0.15	0.1	0.1017	0.1130	0.0000	0.7159
ImageNet	0.15	0.2	0.0979	0.1223	0.0000	0.6626
ImageNet	0.15	0.5	0.0683	0.1367	0.0000	0.4330
ImageNetV2	0.05	0.1	0.0445	0.0494	0.1840	0.4140
ImageNetV2	0.05	0.2	0.0393	0.0491	0.0100	0.3750
ImageNetV2	0.05	0.5	0.0244	0.0489	0.0000	0.2372
ImageNetV2	0.10	0.1	0.0863	0.0958	0.0400	0.5091
ImageNetV2	0.10	0.2	0.0783	0.0979	0.0000	0.4681
ImageNetV2	0.10	0.5	0.0499	0.0996	0.0000	0.3010
ImageNetV2	0.15	0.1	0.1197	0.1331	0.0000	0.5731
ImageNetV2	0.15	0.2	0.1126	0.1407	0.0000	0.5330
ImageNetV2	0.15	0.5	0.0739	0.1479	0.0000	0.3465

Table 6: Full online sweep. Final error, AI error, and budget saved are averages over 500 trials. Rows are grouped by (dataset, ε) scenario and separated by light rules.

D Alternative Confidence Bounds

The Hoeffding bound (11) is simple but may be conservative. Any valid one-sided mean upper confidence bound can be substituted. For example, for a fixed λ, an empirical Bernstein bound gives

$$U_m^{\mathrm{EB}}(\lambda;\alpha) = \widehat{L}_m(\lambda) + \sqrt{\frac{2\hat{V}_m(\lambda)\log(2/\alpha)}{m}} + \frac{7\log(2/\alpha)}{3(m-1)}, \tag{24}$$

where $\hat{V}_m(\lambda)$ is the sample variance of $H_{I_s}(\lambda)$.

When the single-price no-correction argument does not apply, the candidate prices can instead be certified simultaneously by a union bound. Over a finite grid Λ, the uniform Hoeffding bound

$$U_m^{\mathrm{unif}}(\lambda;\alpha) = \widehat{L}_m(\lambda) + \sqrt{\frac{\log(|\Lambda|/\alpha)}{2m}} \tag{25}$$

satisfies $\mathbb{P}\{\exists\,\lambda \in \Lambda : L_T(\lambda) > U_m^{\mathrm{unif}}(\lambda;\alpha)\} \le \alpha$.

Remark 1 (Unequal cheap-source costs)**.** *Under equal cheap-source costs the selected source $j_t^\lambda \in \arg\min_j \widehat{h}_t(j)$ does not depend on λ, the routed loss is monotone in the scalar score $\min_j \widehat{h}_t(j)$, and Theorem 1 certifies the whole price continuum with the single-price bound (11) at level α. Under unequal costs the selected source $j_t^\lambda \in \arg\min_j\{c_t(j) + \lambda\widehat{h}_t(j)\}$ varies with λ, so source switching can make $L_T(\lambda)$ non-monotone and the single-point argument no longer applies. One then certifies a finite price grid Λ with the uniform bound (25)—equivalently, replaces α by $\alpha/|\Lambda|$ in (11)—and Corollary 3 bounds the grid size needed to cover the continuum.*

Betting-based confidence sequences can also be used if the calibration stream is reused while expanding the candidate price set.

E Not Very Meaningful Results

The following observation is included only to show that a continuum of prices can be reduced to a finite policy class on a fixed finite dataset. It is relevant only under unequal cheap-source costs, where source selection varies with λ; under the equal-cost setting of the main text Theorem 1 already certifies all prices without it. It is not a main contribution: the resulting confidence penalty may be larger than that of a deliberately chosen price grid, and it does not remove the uniform-certification cost.

E.1 Continuous Prices and Breakpoint Enumeration

For each item t, set $\widehat{h}_t(E) = 0$ and $\widehat{h}_t(j) = \widehat{h}_t(j)$. The action selected by a fixed price is the lower envelope of the $k+1$ lines

$$c_t(a) + \lambda \widehat{h}_t(a), \qquad a \in \{E, 1, \dots, k\}.$$

With a deterministic tie-breaking rule, the action can change only when two of these lines cross.

Corollary 3 (Uniform certification over all prices)**.** *Fix a bounded interval $[0, \lambda_{\max}]$ and include the expert-only policy as an additional candidate. Let Π_λ be the fixed-price policy induced by (6)–(7) on the union of the target and calibration items. The number of distinct induced policies over $\lambda \in [0, \lambda_{\max}]$ is at most*

$$N_{\mathrm{br}} \le 1 + 2(T+m)\binom{k+1}{2}. \tag{26}$$

Therefore the uniform guarantee holds over all $\lambda \in [0, \lambda_{\max}]$ after replacing $|\Lambda|$ in (25) by N_{br}. Under equal cheap-source costs this enumeration is unnecessary: source selection does not depend on λ, L_T is monotone, and Theorem 1 already certifies the whole continuum with the single-price bound (11).

Proof. For a fixed item, the selected action is constant between consecutive crossing points of pairs of lines $c_t(a) + \lambda \widehat{h}_t(a)$. There are at most $\binom{k+1}{2}$ pairwise crossings per item. Across $T+m$ target and calibration items, there are therefore at most

$$B = (T+m)\binom{k+1}{2}$$

crossing values, counting multiplicity. These crossing values partition $[0, \lambda_{\max}]$ into at most $B+1$ open intervals, and there are at most B exact crossing points. Within each open interval the entire vector of decisions on target and calibration items is fixed. At an exact crossing point, deterministic tie-breaking determines a single additional policy, which may or may not coincide with one of the neighboring interval policies. Thus the number of distinct induced policies is at most $2B+1$. Applying Theorem 1 to one representative price from each induced policy gives the result. $\square$

E.2 Explicit Cost Accounting

This accounting is useful for implementation, but it is not central to the statistical guarantee. The main distinction is between deployment cost, development-label cost, and audit or correction cost.

Deployment labeling cost. For a fixed offline price λ, the average deployment cost on the target pool is exactly $C_T(\lambda)$ from (9), that is, $\frac{1}{T}\sum_{t=1}^{T}[D_t^\lambda c_{t,E} + (1-D_t^\lambda)c_{t,j_t^\lambda}]$. This is the cost optimized in (12). If cheap predictions have already been precomputed at no marginal deployment cost, set $c_{t,j}=0$. If calling a model has a per-token, per-image, or per-query cost, include that cost in $c_{t,j}$.

Offline development cost. If m_{train} expert labels are used to train $\widehat{h}$ and m_{cal} expert labels are used to certify the fixed price, the development cost is

$$C^{\text{dev}} = \sum_{i\in I_{\text{train}}} c_{i,E} + \sum_{i\in I_{\text{cal}}} c_{i,E} + \text{cheap-source costs incurred on development items.} \tag{27}$$

This cost may be treated as sunk if the labeled development data already exist, but it should be included when comparing end-to-end data-curation pipelines.

Audit-or-correction cost. In the offline setting, calibration labels play the role that audits play online. If calibration items are part of the target pool and are corrected to expert labels in the released dataset, then there is no duplicate expert cost for those items. The total realized end-to-end cost can be written as

$$C_T^{\text{total}}(\lambda) = \frac{1}{T}\sum_{t=1}^{T}\Big[(1-D_t^\lambda)c_{t,j_t^\lambda} + \mathbf{1}\{D_t^\lambda \text{ or } t\in I_{\text{cal}}\}c_{t,E}\Big] + C^{\text{train}}/T, \tag{28}$$

where C^{train} denotes expert-labeling cost used only for proxy training and not for final target labels. This formula separates expert labels used for final deferral from expert labels used for statistical certification.

Online audit cost. For Algorithm 2, the conditional expected round-t cost under audit-and-correct is

$$\mathbb{E}[C_t \mid \mathcal{G}_t] = D_t + (1-D_t)p_t.$$

This identity isolates the extra expert cost paid to obtain online feedback; it is not a cost-regret guarantee.

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